



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Useful Systems



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can talk about the reasons to use OOP.
- I can discuss common problems in computer system design.
- I can talk about the importance of robust and maintainable systems.

Ask your partner:

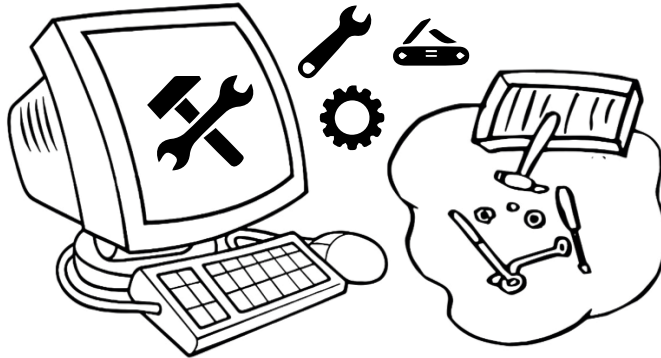
- What do you think are the things that damage a software or system?
- How can OOP improve a common company concern? Why?





Let's Explore!

Two students are getting for a test on OOP.



Instructions: Read the text below.

- Peter:** The test is simple. There are just three main questions about the same topic. Let's start by reviewing common problems in computer system designs.
- Sam:** Wait! I know them. There is a problem regarding poor analysis and design, and there is another regarding maintainability, right?
- Peter:** Great! Now, can those problems affect OOP?
- Sam:** Yes, they can. There are some problems when the system doesn't do the right things, and there are problems when the computer system is hard to understand. In the first case, poor analysis and design are the causes, while poor maintainability is the reason for the second one.
- Peter:** Ok. So, how can OOP reduce maintenance problems?
- Sam:** Well, there is a reason why it is important to use OOP and it is because OOP can help us overcome common problems in computer system design. In other words, software engineers need to make sure the software they design is easy to maintain throughout the years after its initial creation; they can achieve this by using OOP properly. With OOP, they can analyze and design all the tasks for the development of the software during its initial phase and predict its future behavior.
- Peter:** Now that you mention it. Can a system be damaged after its initial development phase?
- Sam:** Sure! They can be caused by the changes that companies do for its development. In this sense, there is an important issue we have to keep in mind, system robustness.
- Peter:** Can you explain why robustness is so important here?
- Sam:** Well, since robustness refers to the ability of a system to cope with errors that can incur during its execution, it is the job of software engineers to guarantee a system is robust and, therefore, maintainable. Through OOP, we can make sure there are systems that are easy to maintain and that are robust. These two features guarantee a system's appropriate functioning.
- Peter:** Well done! I think we're now ready for the test.
- Sam:** Of course we are!



Let's Practice!

Importance of Robustness

Instructions: Proofread the sentences. Correct the mistakes.

1. There are an important issue that we have to keep in mind when talking about systems development, which is system robustness.

2. There is problems that you can avoid with a robust software.

3. When there are a robust software, maintenance is easier to keep.

4. There is some difficulties when there are a different program code.



Let's Practice!

Reasons to Use OOP

Instructions: Select the correct answer to complete the sentences.

1. _____ problems that can incur because of a system's poor analysis and design.
a. There are some
b. There are a
c. There is some
2. _____ many maintenance problems that can be overcome by using some OOP tasks.
a. There's
b. There is
c. There are
3. _____ an initial phase in every software design which is essential for its future behavior.
a. There be
b. There is
c. There are
4. _____ companies that make changes to software systems causing problems. OOP prevents this.
a. There are
b. There is
c. There's
5. _____ problems that can happen but are not a big challenge for robust systems well-designed with OOP.
a. There be
b. There are
c. There is



Extension Activity

A problem to solve!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about one common problem that occurs when engineers design software.
- Write and explain the best way to solve that problem. Is it related to OOP?
- Discuss with your instructor and partners the solution you propose.

Self-Assessment

Yes No

- I research about common problems in the design of software.
- I determine if a software design problem can be solved with OOP.
- I propose a solution for a specific problem regarding computer system design.
- I discuss the solution I propose with my instructor and partners.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- phase** (noun): stage
e.g. With OOP, they can analyze and design all the tasks for the development of the software during its initial phase and predict its future behavior.
- overcome** (verb): to defeat
e.g. Well, there is a reason why it is important to use OOP and it is because OOP can help us overcome common problems in computer system design.
- robustness** (noun): the ability of a system to continue working in spite of execution or input defaults
e.g. In this sense, there is an important issue we have to keep in mind, system robustness.
- maintainability** (noun): ease with which a product or system can be continually restored
e.g. There is a problem regarding poor analysis and design, and there is another regarding maintainability, right?
- incur** (verb): to happen
e.g. Well, since robustness refers to the ability of a system to cope with errors that can incur during its execution, it is the job of software engineers to guarantee a system is robust and, therefore, maintainable.



Learning Points

There is – There are

You use **there is** and **there are** to talk about the existence of something.

You use **there is** and **there are** in simple present tense for affirmative sentences when you want to express the existence of someone or something.

There is is used for singular countable nouns or for uncountable nouns:

***There is** a new software in the company.*

***There is** some time they give us to learn.*

You can use a contraction for this singular form:

There is – There's

There are is used for plural countable nouns:

***There are** issues we have to solve.*

***There are** three specialist who can help us.*

Examples

- *There are just three main questions about the same topic.*
- *There is a problem regarding poor analysis and design, and there is another regarding maintainability, right?*
- *Well, there is a reason why it is important to use OOP and it is because OOP can help us overcome common problems in computer system design.*
- *Through OOP, we can make sure there are systems that are easy to maintain and that are robust.*





Learning Points

Modal can to ask for clarification

You use the modal **can** to ask for clarification about any topic.

Can has different purposes, one of them is to clarify information when you use it in a question. You use the modal **can** for interrogative sentences using the following structure:

Can + subject + verb + complement + ?

For example:

Can you help me with this program?

Can you clarify this concept?

Examples

- *Now, can those problems affect OOP?*
- *Can a system be damaged after its initial development phase?*
- *Can you explain why robustness is so important here?*

